Section 2.6 Limits Involving Infinity

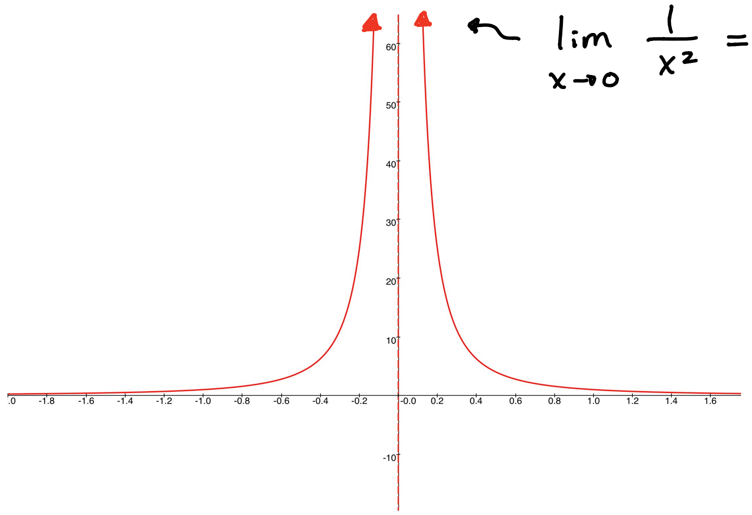
Ex ① Given $f(x) = \frac{1}{x^2}$

A) What happens as x approaches zero?

$$f(0.001) = \frac{1}{0.001^2} = 1{,}000{,}000$$

$$f(-0.001) = \frac{1}{(-0.001)^2} = 1{,}000{,}000$$

As $x \to 0$, $f(x) \to \infty$



$$\lim_{x \to 0} \frac{1}{x^2} = \boxed{\infty}$$

B) What happens as x goes to infinity?

$$f(10000) = \frac{1}{10000^2} = 0.00000001$$

So, as $x \to \infty$, $f(x) \to 0$

$$\lim_{x \to \infty} \frac{1}{x^2} = 0$$

Ex 2

$$\lim_{x\to\infty} \frac{6x^2-7x+1}{3x^2+5x-2}$$

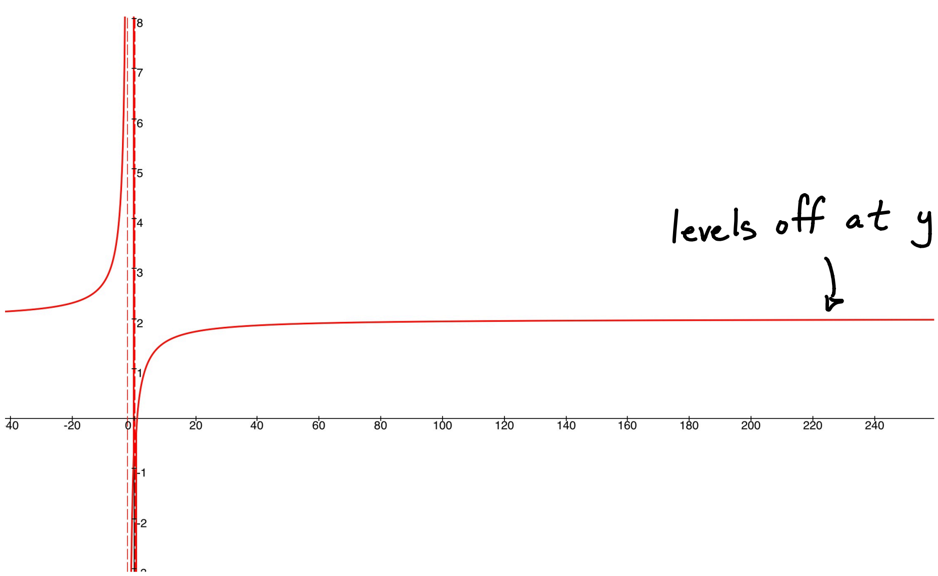
The most powerful term is x^2.

$$= \lim_{x\to\infty} \frac{(6x^2-7x+1)\cdot\frac{1}{x^2}}{(3x^2+5x-2)\cdot\frac{1}{x^2}}$$

$$= \lim_{x\to\infty} \frac{\frac{6x^2}{x^2}-\frac{7x}{x^2}+\frac{1}{x^2}}{\frac{3x^2}{x^2}+\frac{5x}{x^2}-\frac{2}{x^2}}$$

$$= \lim_{x\to\infty} \frac{6-\frac{7}{x}+\frac{1}{x^2}}{3+\frac{5}{x}-\frac{2}{x^2}}$$

$$= \lim_{x\to\infty} \frac{6-\cancelto{0}{\frac{7}{x}}+\cancelto{0}{\frac{1}{x^2}}}{3+\cancelto{0}{\frac{5}{x}}-\cancelto{0}{\frac{2}{x^2}}} = \frac{6}{3} = 2$$



levels off at $y=2$

Ex ③ $$\lim_{x\to\infty} \frac{x^2-16}{(x-4)^2} = \lim_{x\to\infty} \frac{(x-4)(x+4)}{(x-4)(x-4)}$$

$$= \lim_{x\to\infty} \frac{x+4}{x-4} = \lim_{x\to\infty} \frac{\frac{x}{x}+\frac{4}{x}}{\frac{x}{x}-\frac{4}{x}}$$

$$= \lim_{x\to\infty} \frac{1+\cancel{\frac{4}{x}}\nearrow 0}{1-\cancel{\frac{4}{x}}\nearrow 0} = 1$$

Ex ④ $$\lim_{x\to\infty} \frac{5x-8}{x^2-1}$$ divide each term by x^2

$$= \lim_{x\to\infty} \frac{\frac{5x}{x^2}-\frac{8}{x^2}}{\frac{x^2}{x^2}-\frac{1}{x^2}}$$

$$= \lim_{x\to\infty} \frac{\cancel{\frac{5}{x}}\nearrow 0-\cancel{\frac{8}{x^2}}\nearrow 0}{1-\cancel{\frac{1}{x^2}}\nearrow 0} = \frac{0}{1} = \boxed{0}$$

Ex ⑤ $$\lim_{x\to\infty} \frac{\sqrt{x^2+9}}{2x+4} \approx \frac{\sqrt{x^2}}{2x} = \frac{x}{2x} = \frac{1}{2}$$

$\frac{1}{2}$ is my estimate for the answer

$$= \lim_{x\to\infty} \frac{\frac{1}{x}\sqrt{x^2+9}}{\frac{1}{x}(2x+4)}$$ the most powerful term is x

$$= \lim_{x\to\infty} \frac{\sqrt{\frac{x^2}{x^2}+\frac{9}{x^2}}}{2+\frac{4}{x}}$$

$$= \lim_{x\to\infty} \frac{\sqrt{1+\frac{9}{x^2}\to 0}}{2+\frac{4}{x}\to 0} = \frac{\sqrt{1}}{2} = \frac{1}{2}$$

Ex ⑥ $\lim_{x\to\infty} \sqrt{x^2-4} - 2x$ use the conjugate

$$= \lim_{x\to\infty} \left(\sqrt{x^2-4} - 2x\right) \frac{\left(\sqrt{x^2-4}+2x\right)}{\sqrt{x^2-4}+2x}$$

$$= \lim_{x\to\infty} \frac{x^2-4-4x^2}{\sqrt{x^2-4}+2x}$$

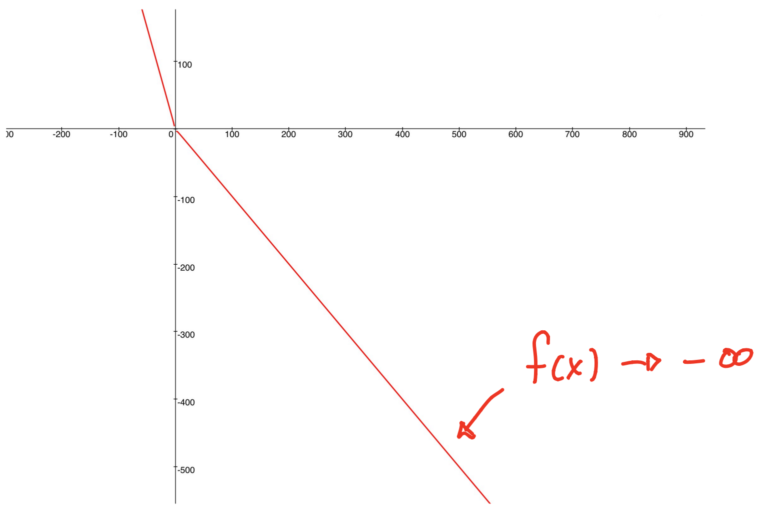
$$= \lim_{x\to\infty} \frac{-3x^2-4}{\sqrt{x^2-4}+2x}$$ the most powerful term is x^2

$$= \lim_{x\to\infty} \frac{\frac{1}{x^2}(-3x^2-4)}{\frac{1}{x^2}(\sqrt{x^2-4}+2x)}$$

$$= \lim_{x\to\infty} \frac{-3-\frac{4}{x^2}}{\sqrt{\frac{x^2}{x^4}-\frac{4}{x^4}}+\frac{2x}{x^2}}$$

$$= \lim_{x \to \infty} \frac{-3 - \frac{4}{x^2}}{\sqrt{\frac{1}{x^2} - \frac{4}{x^4}} + \frac{2}{x}}$$

$$= \lim_{x \to \infty} \frac{-3 - \frac{4}{x^2} \nearrow 0}{\sqrt{\frac{1}{x^2} \nearrow 0 - \frac{4}{x^4} \nearrow 0} + \frac{2}{x} \nearrow 0} \to \frac{-3}{0} \to -\infty$$



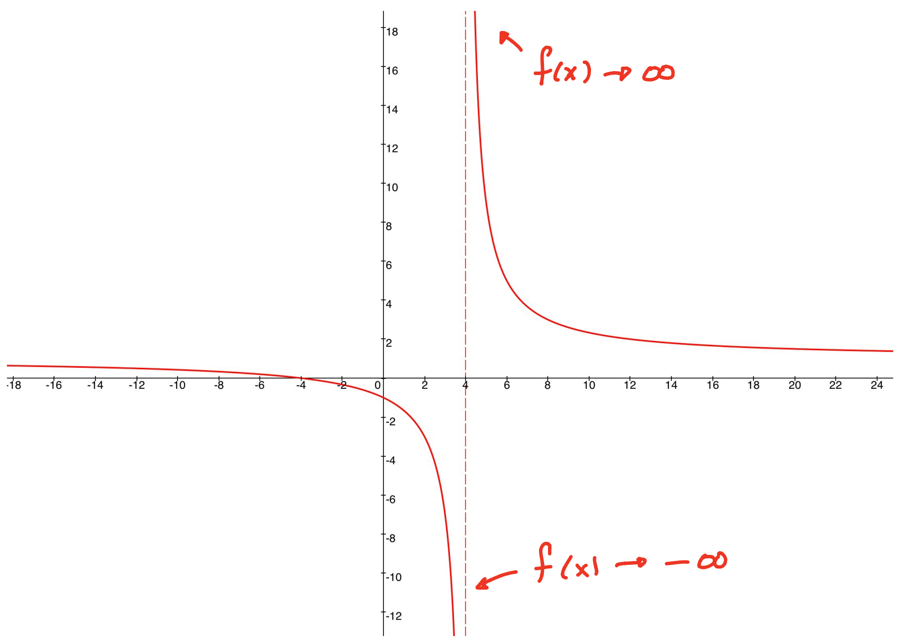
Ex ⑦ $\displaystyle \lim_{x \to \infty} \frac{42}{1 + 5e^{-x}}$

$$= \lim_{x \to \infty} \frac{42}{1 + \frac{5}{e^x} \nearrow 0} = \frac{42}{1} = 42$$

Ex 8 $\displaystyle\lim_{x \to 4} \frac{x^2 - 16}{(x-4)^2}$

$$= \lim_{x \to 4} \frac{(x-4)(x+4)}{(x-4)(x-4)}$$

$$= \lim_{x \to 4} \frac{x+4}{x-4} \to \frac{1}{0}$$



$\displaystyle\lim_{x \to 4} \frac{x^2 - 16}{(x-4)^2}$ does NOT exist since the two sides do NOT agree.